

XpertTeakLite-II Subsystem



For SMT, Oct. 2006

**Highly integrated, low-cost,
low-power platform**

§ Speeds time to market, eases the
integration into customer's SoC

**HW and SW platform provided by a
single source**

§ Reduces risks and complexity

Proven technology

§ Leverages on market adopted technology



Outline

- § **About Us**
- § **CEVA-XpertTeakLite-II Solution**
 - § XpertTeakLite-II building blocks
 - § DSP – TeakLite-II core
 - § DSP subsystem overview
 - § Development Tools
- § **Summary**



About US

- § **CEVA - Semiconductor Intellectual Property (IP) company**
 - § License revenue, royalties on per-shipped chip, on-going maintenance/service fees
 - § Publicly traded (NASDAQ:CEVA, LSE:CVA)

- § **CEVA - The #1 licensor of DSP cores**
 - § Widely used in cellular, DVD, disk drives and consumer electronics
 - § More than 130M chips shipped with CEVA cores in 2005

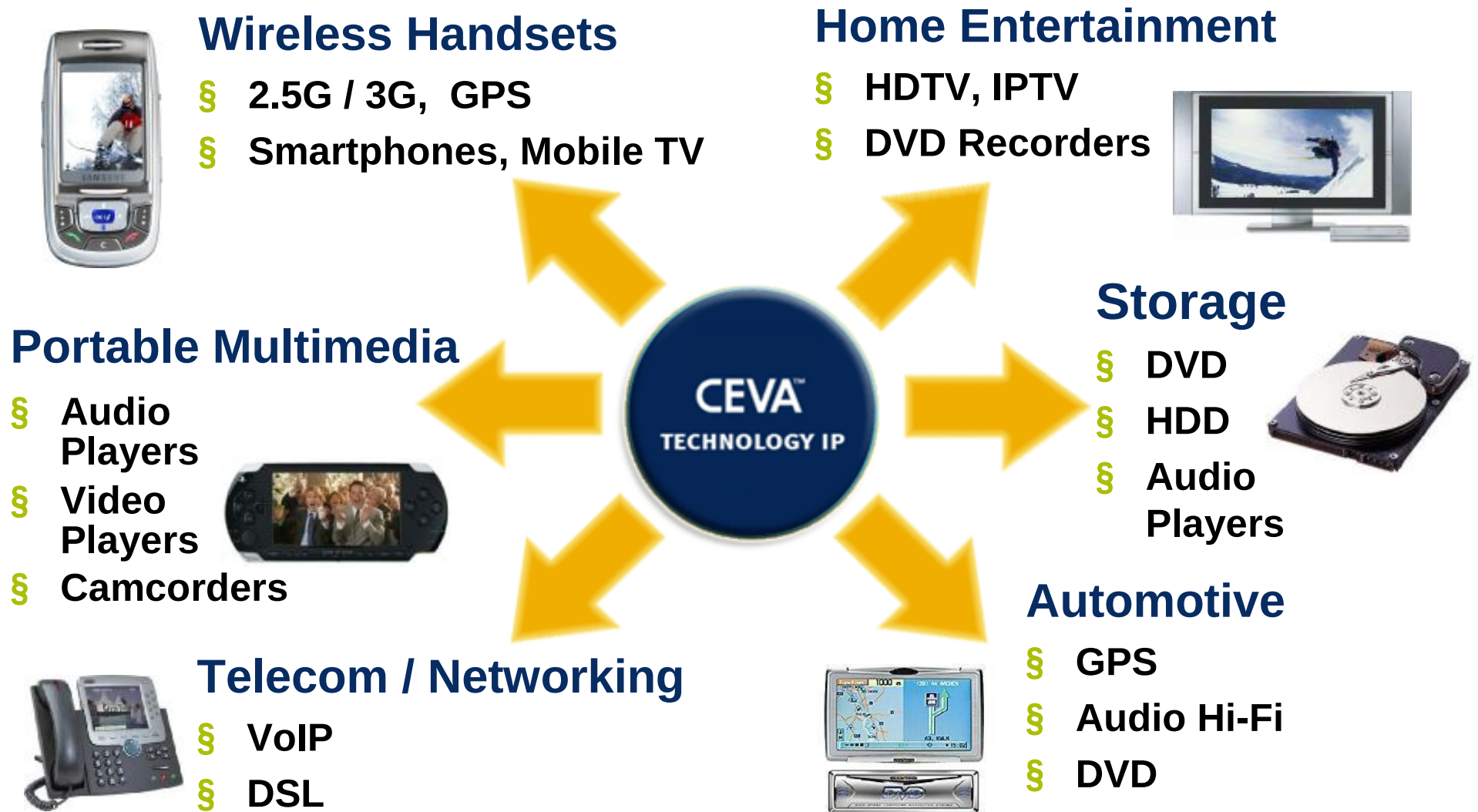
- § **CEVA - Leading products in performance and integration level**
 - § DSP cores – full portfolio of capabilities and products
 - § Platforms - Video, audio, VoIP, Communication and storage

CEVA Licensees

Adopted By World's Top Semiconductors and OEMs

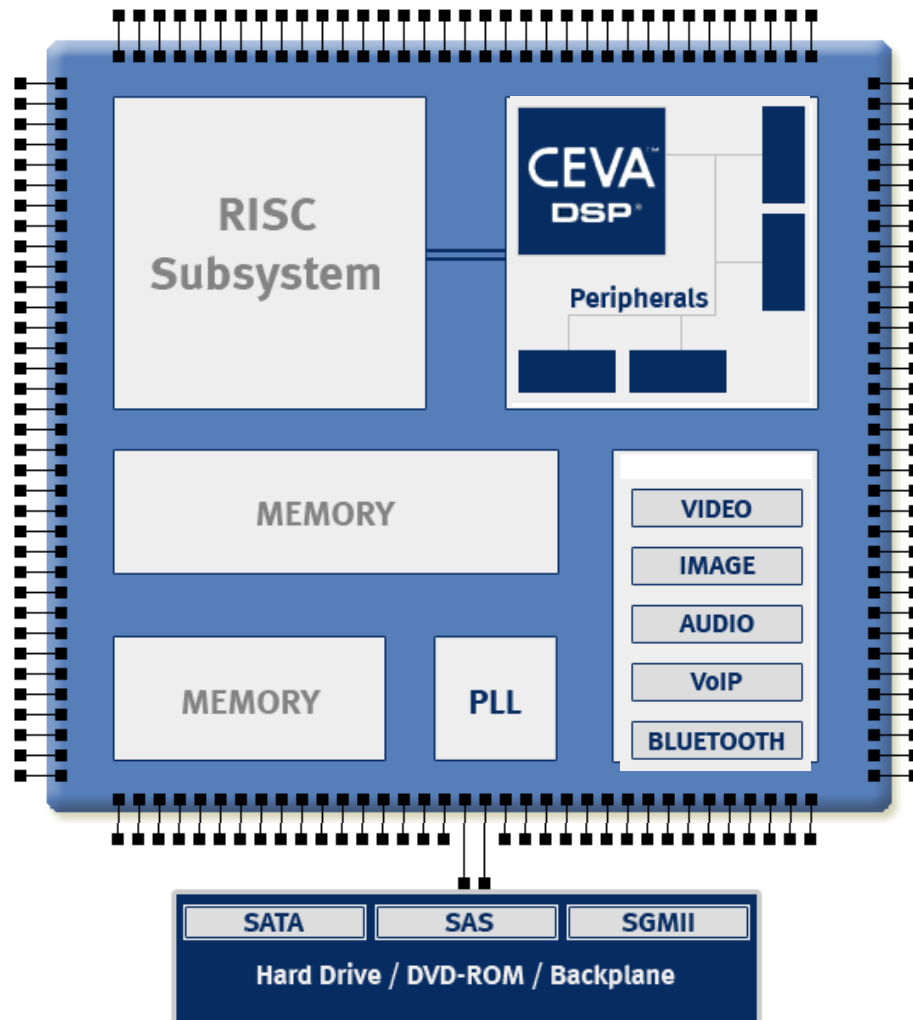


CEVA Addressable Markets



CEVA Product Portfolio

Cores, SoC Platforms, Application platforms



DSP Cores – Signal Processing

CEVA-X1620, CEVA-X1622,
CEVA-Teak, CEVA-TeakLite,
CEVA-TeakLite-II, TeakLite-III

DSP System Platforms

CEVA-XS1200, CEVA-XS1100,
CEVA-XS1102, Xpert-Teak,
Xpert-TeakLite-II

Application Platforms

Video, Audio, Image, VoIP,
Bluetooth

High-speed Serial Storage

SATA 1.5/3.0 Gbps, SAS, SGMII



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§ XpertTeakLite-II building blocks

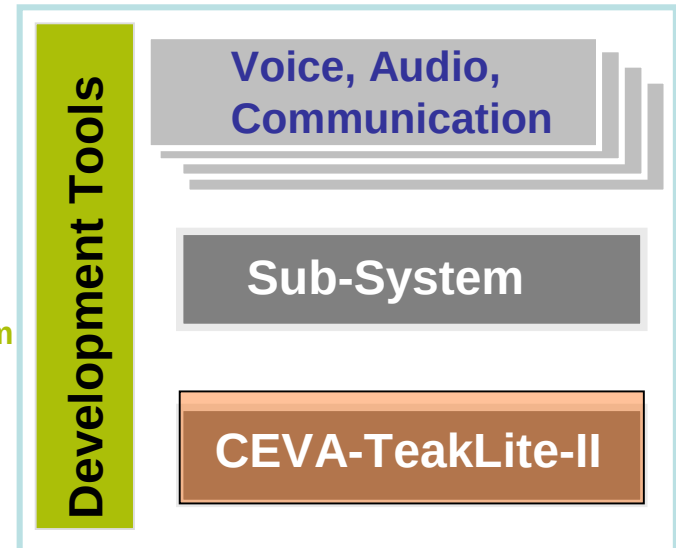
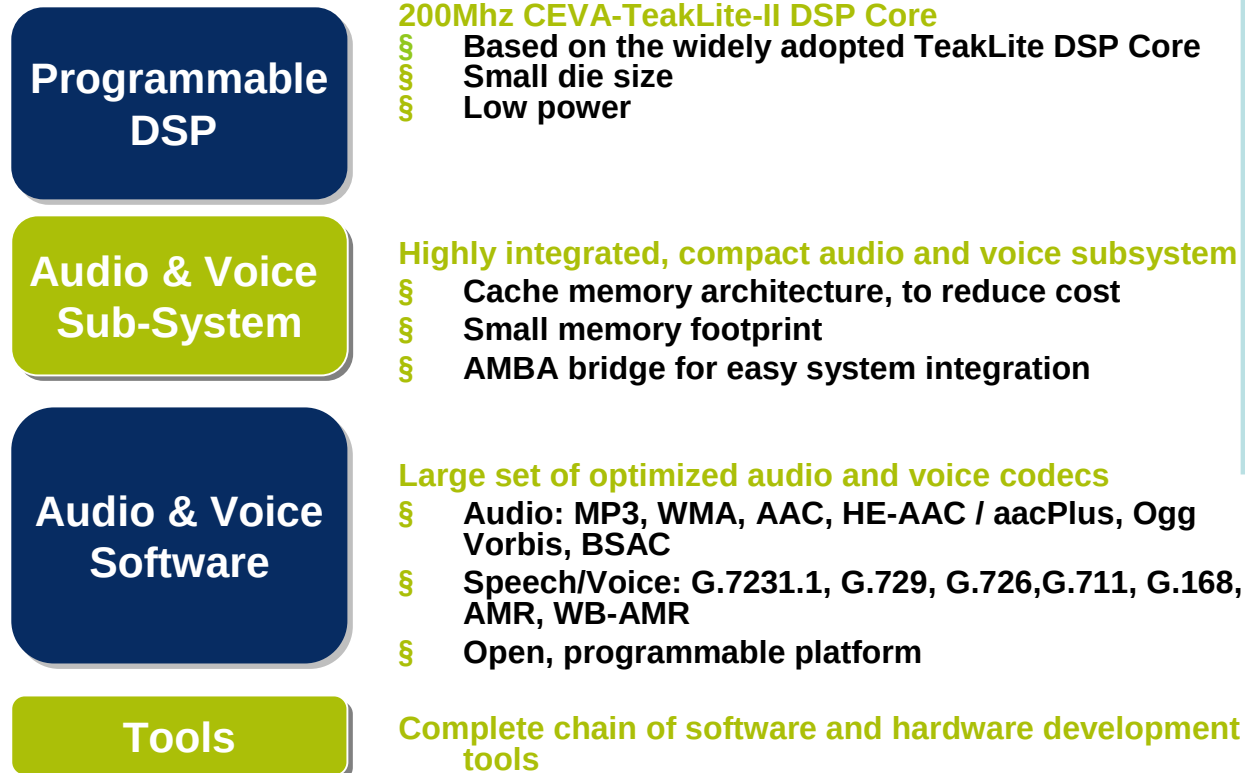
§ DSP – TeakLite-II core

§ DSP subsystem overview

§ Development Tools

§ Summary

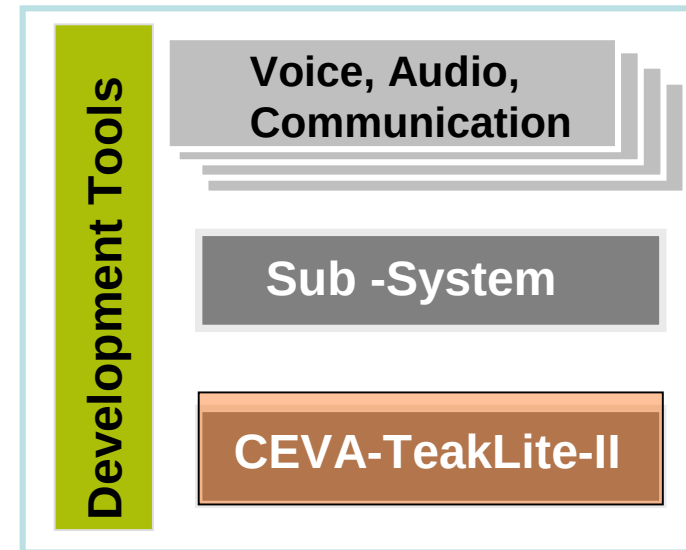
XpertTeakLite-II The building blocks



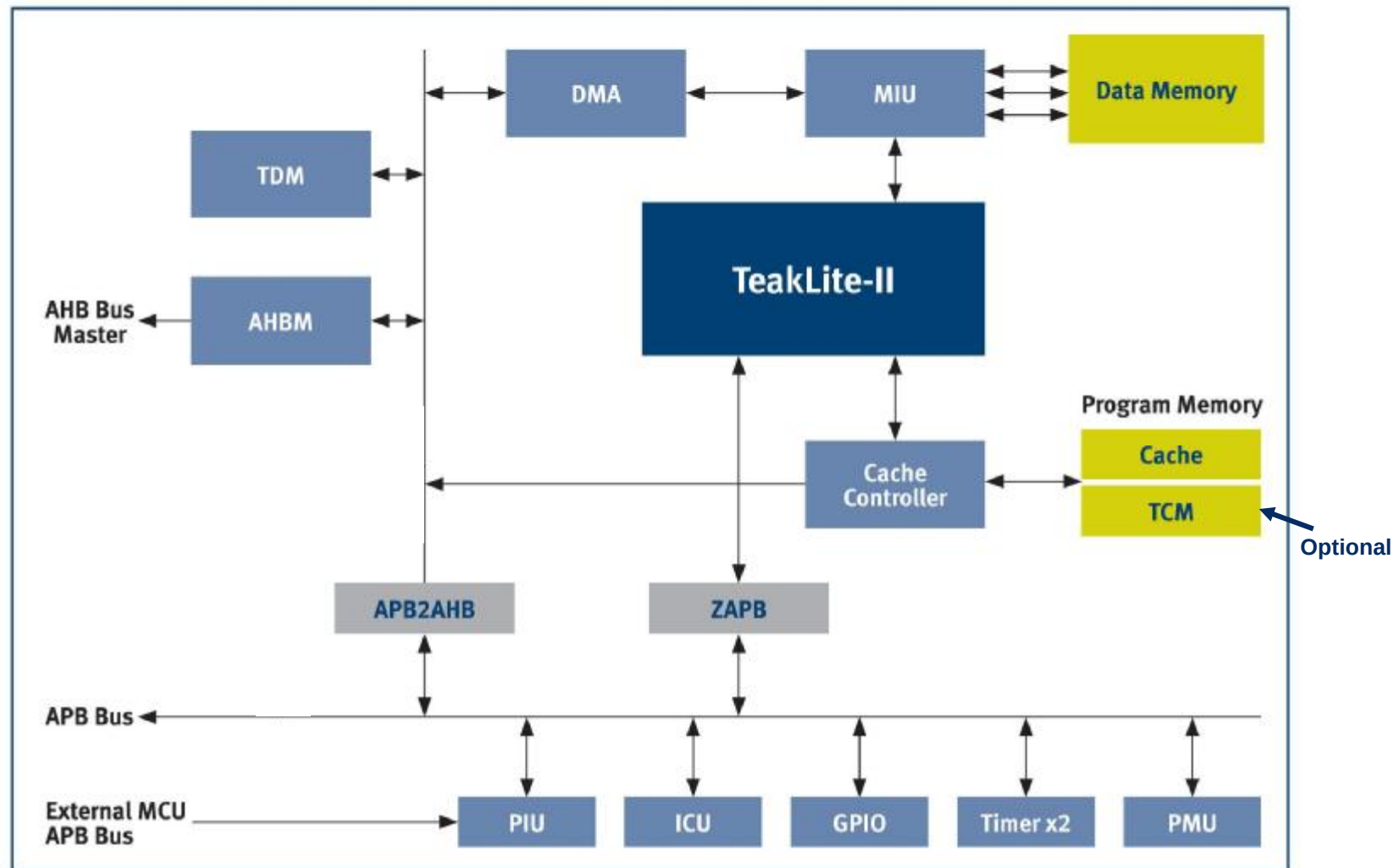
Single source, a complete and highly-integrated platform

DSP Core - TeakLite-II

- § **Fully programmable DSP Core**
 - § Mixture of DSP and Control instructions
- § **High performance**
 - § Up to 200MHz @ 0.13u
 - § 0.5 mm² @ 0.13u
- § **Very low-cost and low-power DSP**
 - § High code density – 16-bit instructions width
- § **Binary-compatible to TeakLite architecture**



Block Diagram



DSP Subsystem Overview

Highly integrated, highly efficient DSP Subsystem

§ Cache memory

- § Low cost solution

§ 3 channel DMA

- § Reduce load from the DSP

§ Ethernet MAC with MII (RMII, SMII, GMII) interface

§ External interfaces

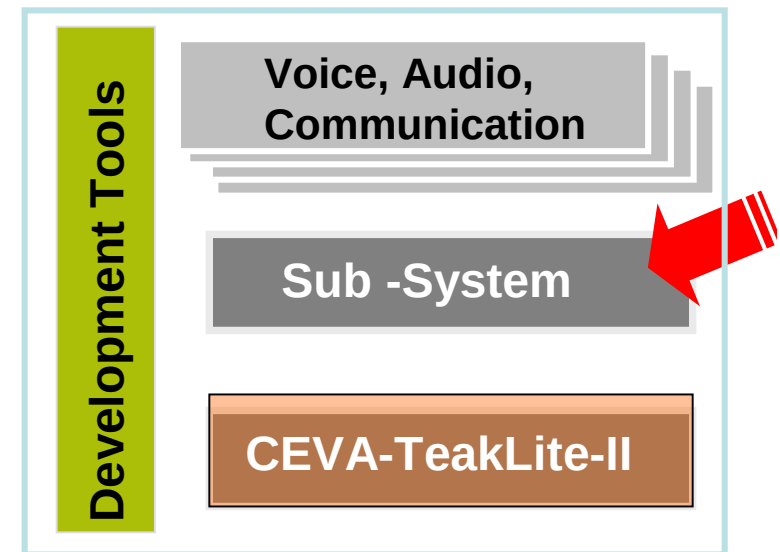
- § AHB Master interface
- § APB to add off the shelf APB peripherals
- § APB slave port for inter processor communications

§ APB peripherals

- § Power Management Unit, Timers, GPIO, Interrupt controller, Processor Interface Unit
- § 2 TDM modules (codec interface)

§ Configurable

- § Data memory – 8KW / 12KW / 16KWord / 32KWord
- § Program memory
 - § Cache – 2KW / 8KW
 - § Tightly coupled memory (TCM), up to 2KWord



Memory Access and DMA

§ Program Memory

- § 2K word cache
- § Optional 2K word Tightly Coupled Memory (TCM)
 - § Under programmer control

§ Data Memory

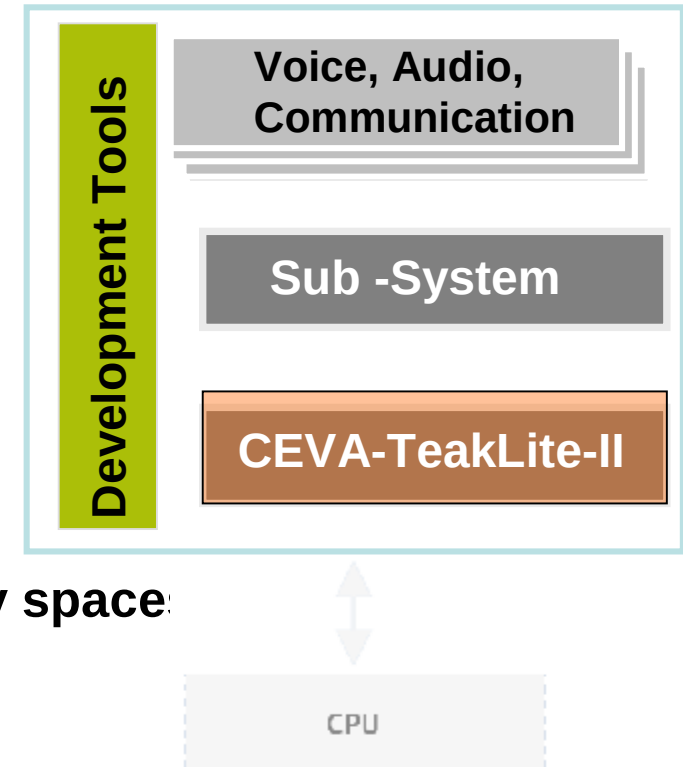
- § 4K word X memory
- § 4K word Y memory

§ Direct access to all data and program memory space:

- § X & Y spaces
- § AHB
- § BTDMP

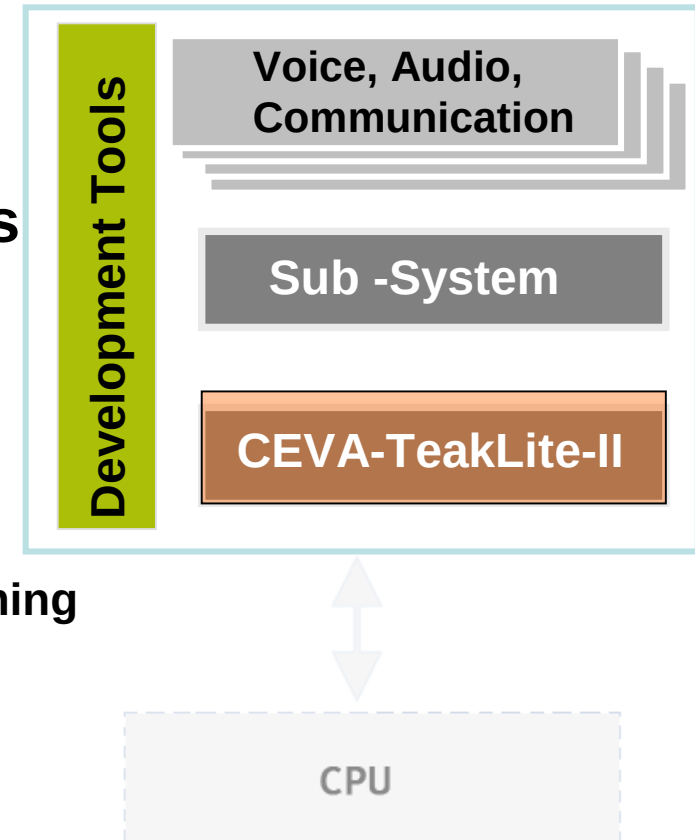
§ Three fully software programmable DMA channels

- § Each channel is fully programmable
- § Simultaneous DSP and DMA Memory Access
- § Emulation support



➤ Power Management

- § **Power Management Unit drives the clocks to all XpertTeakLite-II modules**
 - § Supports clock division
- § **Low power modes**
 - § **Slow clock mode**
 - § The clock is divided by software programming
 - § **Stop clock modes**
 - § Selective shut down of modules
 - § Over-all shut down
 - § External Wake-up input



Interface to External World

§ Program Interface Unit (PIU)

- § Command/Reply protocol registers for host communication
- § Semaphore mechanism
- § Interrupt generation for host and DSP to APB interfaces by:
 - § DSP
 - § Host
- § Asynchronous APB interface (clock decoupling)

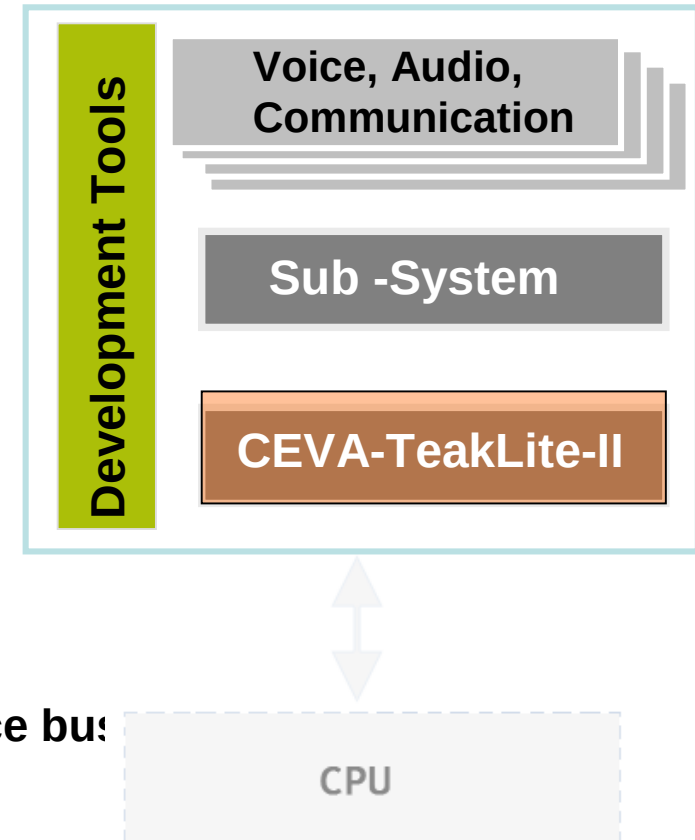
§ ASHB Master Bridge

- § Implements AMBA-compliant high-performance bus controller

§ External APB Master Connection

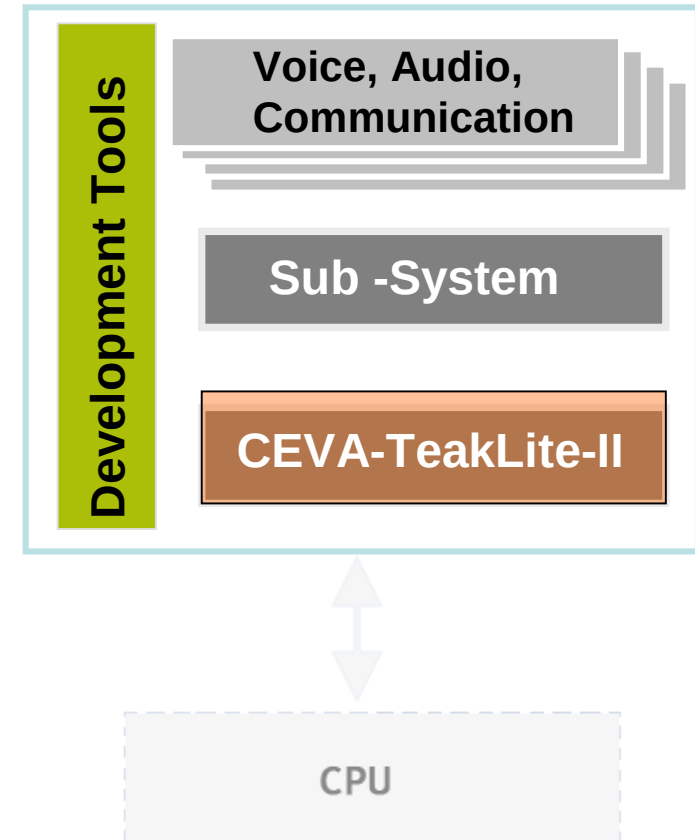
- § Up to two external APB device can be connected

§ 16 General Purpose I/Os (GPIO)



Other Peripherals

- § **Interrupt Control Unit (ICU):**
16 interrupt sources, each can be triggered by
 - § External input
 - § Internal event
 - § Software
 - § Edge/Level interrupts (configurable)
- § **Timers**
2 Timers, supporting various running mode
 - § Selective source – external or internal clock
 - § Cascade mode
 - § Interrupt generation
 - § PWM output
 - § Breakpoint support
 - § Watchdog Timer

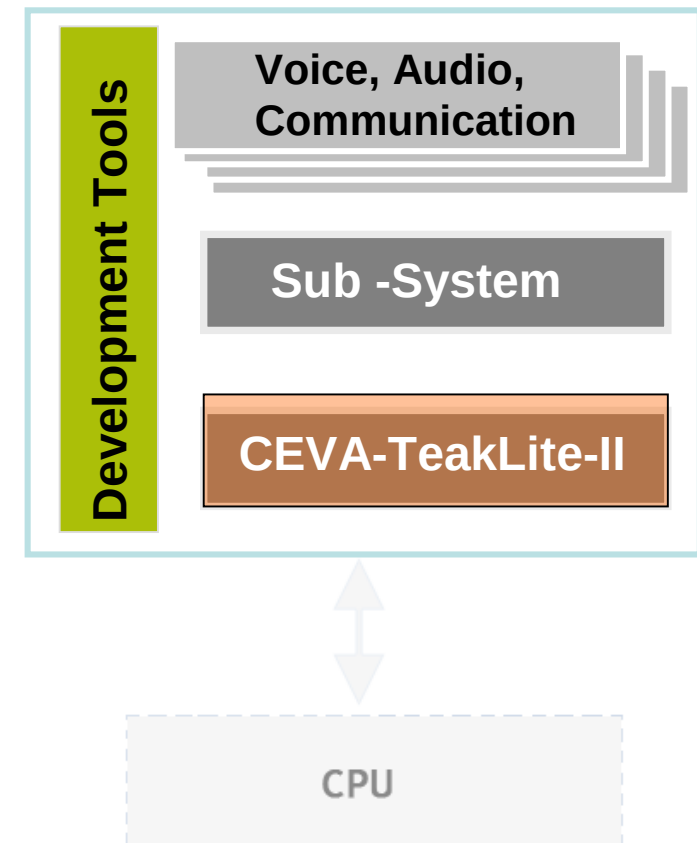


Optional Peripheral

§ Buffered Time Domain Multiplexing Ports

(BTDMP)

- § 1 channel Rx, 1 channel Tx
- § Full-duplex, bi-directional communication with external serial devices
 - § CODECS
 - § analog interface chips
 - § A/D and D/A converters
- § Compatible to variety of standards
 - § PRI, E1, T1, IOM-2, I2S, AC97



CEVA-SDT Tools

Mature and completely proven set of tools

§ Complete and Integrated Software Development Tools

§ Advanced User Friendly IDE

§ Optimized C/C++ Compiler

§ Assembler, Linker & Utilities

§ Advanced GUI Debugger

§ MATLAB connection

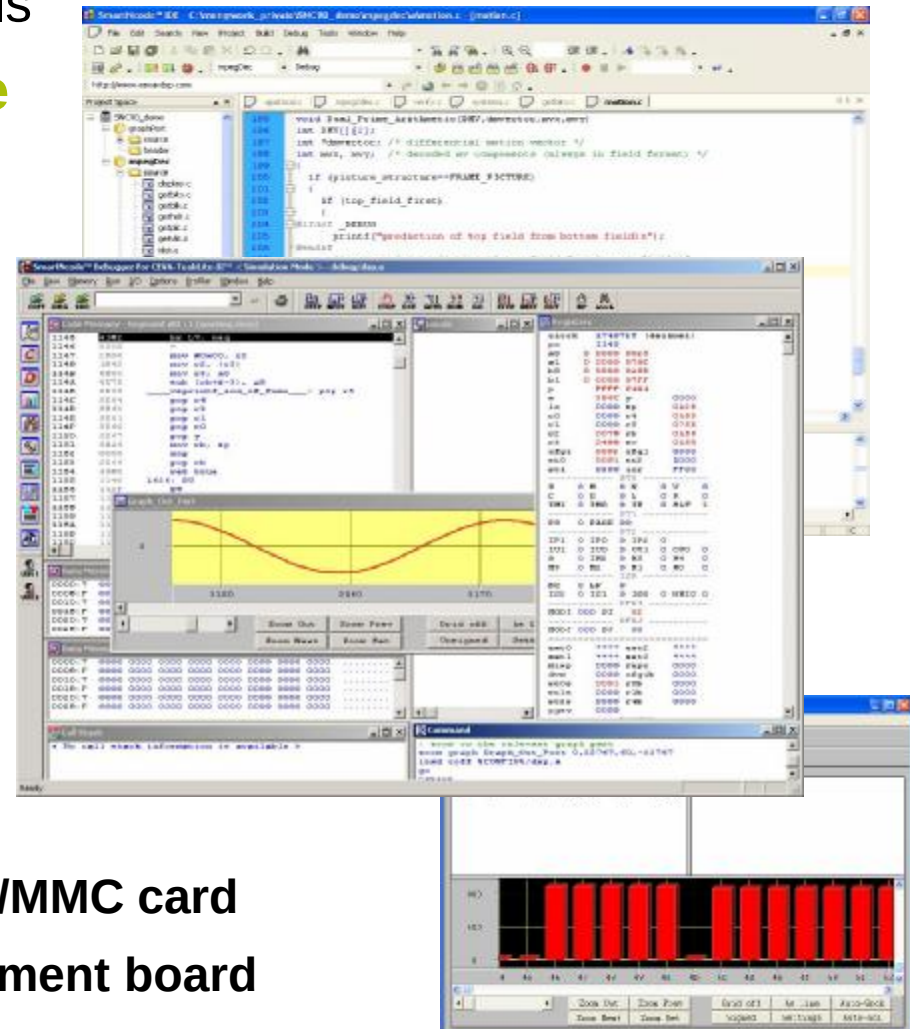
§ Application Profiler

§ Performance Measurement

§ System Boards

§ Support for audio-in, audio-out, SD/MMC card

§ Easy integration with ARM development board





CEVA-SDT Tools

- § **In-house development of advanced software development environments for the last 15 years**
 - § **CEVA recognizes SDT importance for it's CEVA-DSPs – #1 DSP selection criteria according to “Forward Concepts”**
 - § **Knowledgeable support – tools developers support the tools**
 - § **Fast response for customer requests – CEVA's customers are 1st (and only), no need to engage with 3rd parties (SOW, contracts,...)**
 - § **Focus on CEVA's customers needs – we support only CEVA's DSPs**
 - § **Affecting CEVA-DSPs architectures to be compliable**
 - § **Annual new releases including new features and bug fixes**
- § **Mature and completely proven set of tools for all the cores**
- § **Available on PC/Windows and UNIX/Solaris/Linux**

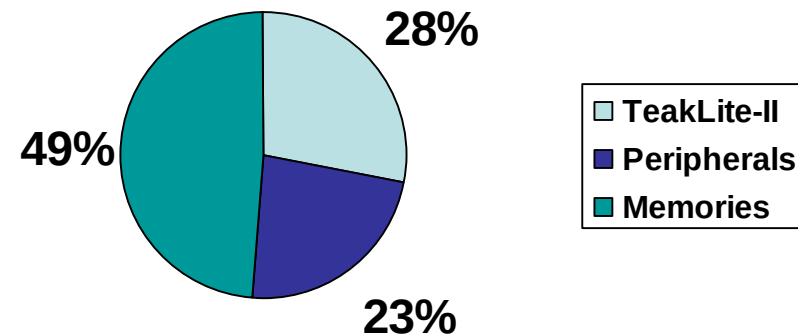


SmartNcode™
Software Development Tools

XpertTeakLite-II

Gate Count and Area (90nm)

Block	Gate Count (K-gates)	Area (mm^2)
Teaklite-II (include Ocem)	75	0.241
MIU	3.5	0.011
DMA 3 Channels	12	0.039
ICU	1.5	0.005
GPIO	1	0.003
PMU	1	0.003
Z2APB	1	0.003
AHBM	10	0.032
Cache-Controller	2.5	0.008
Timer X 2	4.5	0.014
SIC	4.5	0.014
APB2AHB	1	0.003
PIU	3	0.010
BTDMF Tx	7.5	0.024
BTDMF Rx	7.5	0.024
Program 2K cache		0.078
Program Tag (4 instructions)		0.025
Data Memory 2K (A)		0.078
Data Memory 2K (B)		0.078
Data Memory 4K (C)		0.150
Total:	135.5	0.99



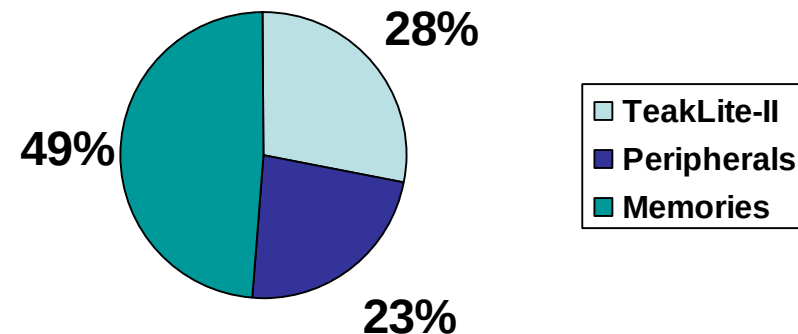
Notes:

1. TSMC 90nm/G, Artisan Library
2. Total area includes place and route overhead (assuming respectively 75% / 85% for logic / top levels)

XpertTeakLite-II

Gate Count and Area (0.13um)

Block	Gate Count (K-gates)	Area (mm^2)
Teaklite-II (include Ocem)	75	0.51
MIU	3.5	0.02
DMA 3 Channels	12	0.08
ICU	1.5	0.01
GPIO	1	0.01
PMU	1	0.01
Z2APB	1	0.01
AHBM	10	0.07
Cache-Controller	2.5	0.02
Timer X 2	4.5	0.03
SIC	4.5	0.03
APB2AHB	1	0.01
PIU	3	0.02
BTDMP RX	7.5	0.05
BTDMP TX	7.5	0.05
Program 2K cache		0.17
Program Tag (4 instructions)		0.07
Data Memory 2K (A)		0.17
Data Memory 2K (B)		0.17
Data Memory 4K (C)		0.29
Total:	135.5	2.11



Notes:

1. TSMC 0.13u/G, Artisan Library
2. Total area includes place and route overhead (assuming respectively 75% / 85% for logic / top levels)



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Area

Less than 1 mm² including local memories *

§ DSP Core engine – CEVA-TeakLite-II

- § Fully programmable DSP Core - only 0.24 mm²
- § 16-bit instruction set reduces program memory size

§ Cache memory subsystem

- § Small on-chip memory area

§ Efficient and small area peripherals tuned for audio platform

Low Cost Platform

* In 90nm TSMC G process, basic configuration



Power Consumption

0.06mA/Mhz for stereo MP3 decoder

- § Built-in PMU (Power Management Unit)
- § Low power TeakLite-II DSP Core
- § Low power audio subsystem
- § Clock gating

§ Figures are in 0.13u TSMC G process



Low Power Consumption Platform

CEVA-XpertTeakLite-II : Key Benefits

- § **Integrated, programmable audio platform**
 - § DSP, subsystem and audio codecs
 - § Short time to market
 - § Low risk
- § **Meets market requirements**
 - § Low cost: 1 mm² including DSP, memory and peripherals (TSMC 90nmG)
2 mm² including DSP, memory and peripherals (TSMC 0.13G)
 - § Low power: 0.06 mW/MHz for stereo MP3 decoder (TSMC 90nmG)
0.1 mW/MHz for stereo MP3 decoder (TSMC 0.13G)
 - § Large set of audio codecs
- § **Proven technology**
 - § Leverages on widely-deployed TeakLite technology
 - § Embedded in over 500 million chips
- § **One-stop-shop**
 - § Solution provided by a single source
 - § Reduces risk and complexity

CEVA-XpertTeakLite-II : High Quality IP

CEVA Verification: stringent functional verification

- § Simulation-proven via systematic tests & Specman
- § Tested real applications in FPGA environment
- § DbgVerifier: Pre and post-silicon debugging functionality
 - § Also available for customer systems

Customer Integration:

- § Robust RTL to GDSII design flow using mainstream, industry-standard EDA tools
- § Run-time Evaluation and Development Platform (EDP)

Customer Simulation:

- § Complete environment to allow verification of integration into customer's SoC
- § Fully synthesizable RTL code

Development & Debug Tools: complete development tool set

- § Compiler, debugger, linker, IDE
- § Software Codecs & DSP Libraries
- § Hardware Development Platform

Support:

- § Single source for all materials = single source of 24x7 support



End

Xpert TeakLite-II

Thank You...

Web: www.ceva-dsp.com